

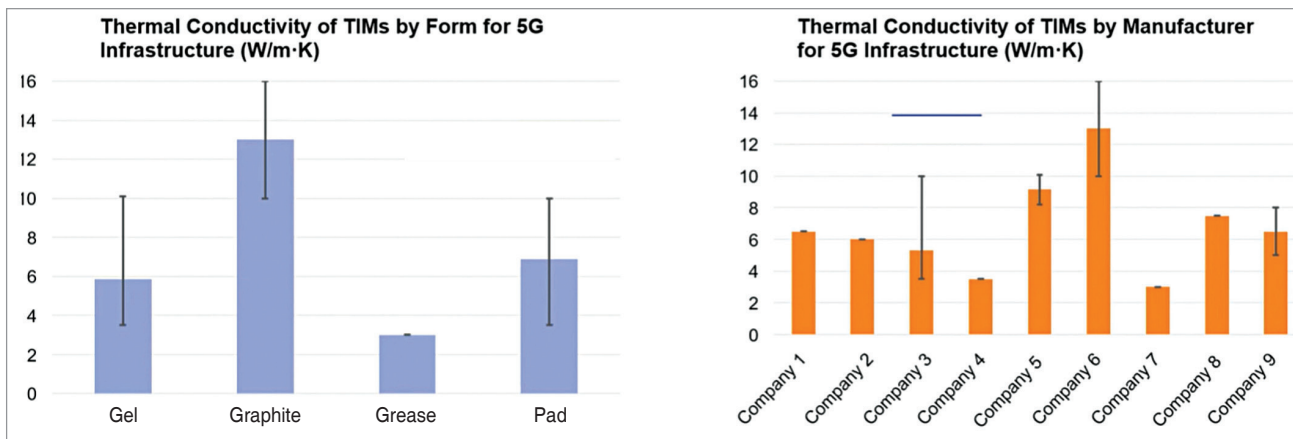


Fig. 3: Average thermal conductivity of TIMs advertised for 5G infrastructure with range of products indicated (error bars)

infrastructure, where thermal management is critical due to the high heat output of these systems.

In data centres, TIMs manage heat generated by high-performance servers and switchboards. The trend is towards higher thermal conductivity to meet the demands of advanced computing and artificial intelligence applications. For instance, TIM areas per data centre switchboard are carefully analysed to ensure efficient cooling, with the average TIM area for data centre switches summarised across different models and supervisory levels.

In the 5G market, the thermal conductivity requirements for TIMs are notably higher, often exceeding 5W/mK. This is necessary to manage the substantial heat output

of 5G infrastructure components. Various TIM forms, such as graphite, grease, and pads, are used, with typical thermal conductivities ranging between 4W/mK and 6W/mK, and even higher for specialised applications.

In the ADAS sector, TIMs are integral to the thermal management of such sensors as cameras, radars, and lidars. These sensors, crucial for autonomous driving, require TIMs with high thermal conductivity to maintain performance and reliability. For example:

Cameras. Require TIMs with thermal conductivity between 1W/mK and 4W/mK, potentially increasing for higher-resolution sensors.

Radars and LIDARs. Typically require TIMs with thermal

conductivity between 3W/mK and 5W/mK, with future needs exceeding 5W/mK due to technological advancements.

As demand for 5G technology continues to rise, TIMs are poised to become critical in the development and performance of next-generation electronic devices. One of the key aspects of 5G applications is electromagnetic interference (EMI) shielding, which is a key growth driver for TIMs. The transition from 4G to 5G presents new challenges in thermal management, especially with higher frequencies and increased equipment density required for 5G networks.

The role in shielding in 5G

5G technology operates at frequencies used by 4G and mid-range frequencies, but it also expands into ultra-high frequencies ranging from 25GHz to 50GHz. This requires more network equipment and base stations to deliver stable signals, resulting in a substantial increase in micro-antennas and base stations compared to 4G technology. Micro-antennas are smaller and more numerous, driving up the volume of TIMs used to manage the heat generated by these densely packed components.

As 5G technology advances,

Tips for choosing the perfect thermal interface materials

Several critical factors must be meticulously considered in large-scale applications, such as automotive and consumer electronics, to ensure optimal performance and cost-effectiveness.

- **Cost.** This is particularly important for large-scale applications, such as automotive and consumer electronics, where budget constraints play a significant role.
- **Thermal conductivity.** Materials with high thermal conductivity, such as boron nitride and carbon nanotubes, are highly effective but tend to be more expensive.
- **Electrical conductivity.** While materials like graphene and carbon nanotubes excel in thermal conductivity, their electrical properties must be carefully managed to avoid potential issues.

By navigating these complexities, engineers and designers can create innovative solutions that meet performance requirements and align with economic objectives, driving progress in these dynamic sectors.